Assignment 6 — Integrity Constraints

Project: College Management System | Database: college_management_system | SQL dialect: MySQL 8.0

Based on the seven-table college design supplied for this project. Example records and expected outputs are illustrative, not screenshots from MySQL Workbench.

Implemented constraints

PRIMARY KEY uniquely identifies each record; FOREIGN KEY prevents orphan references; UNIQUE prevents duplicate email addresses, repeated registrations and repeated exam results; CHECK limits credits, semester, marks and exam type; DEFAULT CURRENT_DATE supplies an enrollment date if omitted.

Constraint SQL

```
CREATE TABLE Department (Department_ID INT PRIMARY KEY, Department_Name VARCHAR(100) NOT
NULL UNIQUE, Location VARCHAR(100) NOT NULL, HOD VARCHAR(100));
CREATE TABLE Student (Student_ID INT PRIMARY KEY, Name VARCHAR(100) NOT NULL, Email
VARCHAR(100) NOT NULL UNIQUE, Phone VARCHAR(15), Department_ID INT NOT NULL REFERENCES
Department(Department_ID), Admission_Year INT NOT NULL CHECK(Admission_Year BETWEEN 2000
AND 2100));
CREATE TABLE Course (Course_ID INT PRIMARY KEY, Course_Name VARCHAR(100) NOT NULL, Credits
INT NOT NULL CHECK(Credits BETWEEN 1 AND 6), Department_ID INT NOT NULL REFERENCES
Department(Department_ID), UNIQUE(Course_Name,Department_ID));
CREATE TABLE Faculty (Faculty_ID INT PRIMARY KEY, Name VARCHAR(100) NOT NULL, Email
VARCHAR(100) NOT NULL UNIQUE, Phone VARCHAR(15), Department_ID INT NOT NULL REFERENCES
Department(Department_ID));
CREATE TABLE Enrollment (Enrollment_ID INT PRIMARY KEY, Student_ID INT NOT NULL REFERENCES
Student(Student_ID), Course_ID INT NOT NULL REFERENCES Course(Course_ID), Semester INT NOT
NULL CHECK(Semester BETWEEN 1 AND 8), Enrollment_Date DATE NOT NULL DEFAULT (CURRENT_DATE),
UNIQUE(Student_ID,Course_ID,Semester));
CREATE TABLE Examination (Exam_ID INT PRIMARY KEY, Course_ID INT NOT NULL REFERENCES
Course(Course_ID), Exam_Date DATE NOT NULL, Exam_Type VARCHAR(50) NOT NULL CHECK(Exam_Type
IN ('Midterm','Final','Quiz')), Total_Marks INT NOT NULL CHECK(Total_Marks > 0));
CREATE TABLE Result (Result_ID INT PRIMARY KEY, Exam_ID INT NOT NULL REFERENCES
Examination(Exam_ID), Student_ID INT NOT NULL REFERENCES Student(Student_ID), Marks INT NOT
NULL CHECK(Marks >= 0), Grade VARCHAR(5) NOT NULL CHECK(Grade IN ('A+','A','B','C','D',
'F')), UNIQUE(Exam_ID,Student_ID));
```

Valid verification statements

```
SELECT COUNT(*) AS Student_Count FROM Student;
SELECT COUNT(*) AS Enrollment_Count FROM Enrollment;
SELECT COUNT(*) AS Result_Count FROM Result;
```

Expected counts

Table	Rows
Student	6
Enrollment	9
Result	9

Negative tests — run individually; errors are expected

```
-- Duplicate student email: should fail UNIQUE
INSERT INTO Student VALUES (999,'Test','aarav@college.edu','999',1,2026);
-- Unknown department: should fail FOREIGN KEY
INSERT INTO Student VALUES (998,'Test','test998@college.edu','999',999,2026);
-- Invalid semester: should fail CHECK
INSERT INTO Enrollment VALUES (999,101,201,10,'2026-09-01');
```

Run these three invalid INSERT commands separately. They are expected to be rejected, so do not include them in the main successful execution script. A Result.Marks upper bound relative to Examination.Total_Marks requires a trigger or application-level validation because a standard CHECK cannot reference another table.